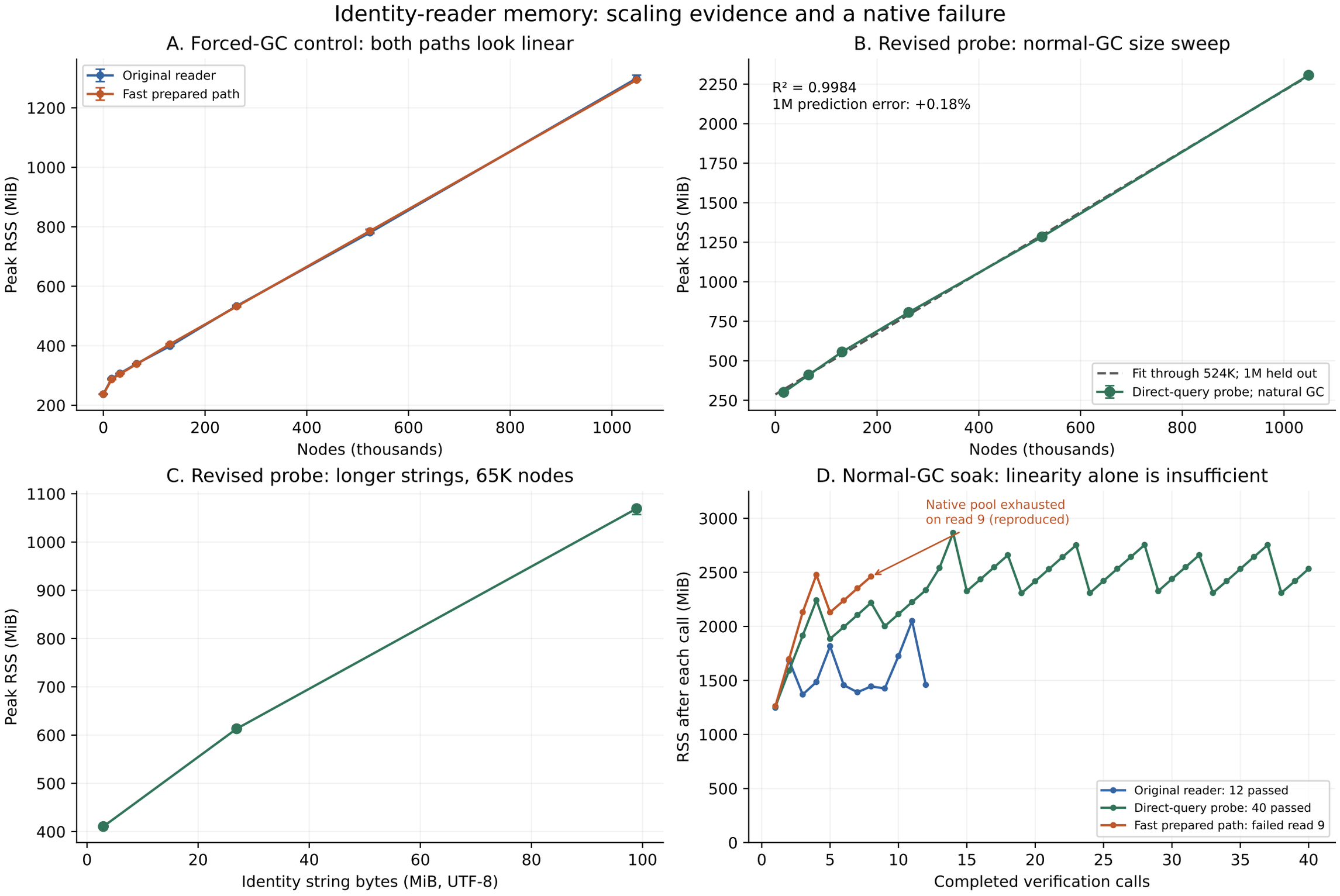
Identity-reader memory: scaling evidence and a native failure
A. Forced-GC control: both paths look linear
Original reader
Fast prepared path
Peak RSS (MiB)
Nodes (thousands)
B. Revised probe: normal-GC size sweep
R² = 0.9984
1M prediction error: +0.18%
Fit through 524K; 1M held out
Direct-query probe; natural GC
Peak RSS (MiB)
Nodes (thousands)
C. Revised probe: longer strings, 65K nodes
Peak RSS (MiB)
Identity string bytes (MiB, UTF-8)
D. Normal-GC soak: linearity alone is insufficient
Native pool exhausted
on read 9 (reproduced)
Original reader: 12 passed
Direct-query probe: 40 passed
Fast prepared path: failed read 9
RSS after each call (MiB)
Completed verification calls
A–C: median and range of 3 fresh processes. Fixed 512 MiB database pool; Node 22.23.3 / LadybugDB 0.22.0.
A forces GC between reads; B–D use natural GC. No production implementation or universal memory guarantee.